Simplex Algorithm

Goal is to find an optimal solution starting at a basic feasible solution

Def Simplex Multipliers mrt B

$$\pi = (B^{-1})^T c_b$$

Def Reduced Costs mrt B

$$c^\pi = c - A^T\pi$$

Note that $c^\pi = (c_B, c_L) - (B^T, L^T)\pi$

$$\to (c_B, c_L) - (B^T, L^T)(B^{-1})^T c_B$$

$$\to (c_B - B^T(B^{-1})^T c_B, c_L - L^T(B^{-1})^T c_B)$$

$\to (0, c_L - L^T(B^{-1})^T c_B)$ (Note that $B^T(B^{-1})^T = (B^{-1}B)^T = I$)

For finding equivalent points, $(c^\pi)^T x = c^T x - \pi^T A x = c^T x - \pi^T b$ (note $\pi^T b$ is a constant)

This is because $\min c^T x = (c^\pi)^T x \equiv \min (c^\pi)^T x$

Main Idea:

$c^\pi_j < 0 \to increasex_j, updatex_B$ and repeat until optimal solution (no $c^\pi_j < 0$)

Small issue is that a change of basis doesn't necessarily change extreme points